

Roots (Part IV): Negative Radicands, Imaginary Numbers, and Complex Numbers (Homework)

Simplify. *Do not assume that the unknowns are positive.*

① $\sqrt{x^2}$

② $\sqrt[3]{x^3}$

③ $\sqrt[7]{x^{56}}$

④ $\sqrt[4]{x^{12}}$

⑤ $\sqrt[4]{x^{24}}$

⑥ $\sqrt{x^8}$

⑦ $\sqrt{24x^{30}}$

⑧ $\sqrt{64x^{32}}$

⑨ $\sqrt[4]{16x^{36}}$

⑩ $\sqrt[4]{8x^{14}}$

⑪ $(6+3i)+(1-2i)$

⑫ $(-8+4i)+(-2-3i)$

⑬ $(9-i)+(3+3i)$

⑭ $(-6-5i)-(1-2i)$

⑮ $(2+7i)-(1+4i)$

Answers

$|x|$

x

x^8

$|x^3|$

x^6

x^4

$2|x^{15}|/\sqrt{6}$

$8x^{16}$

$2|x^9|$

$|x^3|/\sqrt{8x^2}$

$7+i$

$-10+i$

$12+2i$

$-7-3i$

$1+3i$

⑯ $(-10+3i)-(-2-9i)$

$-8+12i$

⑰ $7i(6i)$

-42

⑱ $-3(6i)$

$-18i$

⑲ $4i(-2)$

$-8i$

⑳ $-10i(-9i)$

-90

㉑ $3i(7+8i)$

$21i-24$

㉒ $-2i(1-6i)$

$-2i-12$

㉓ $i(-5+4i)$

$-5i-4$

㉔ $6i(-2-2i)$

$-12i+12$

㉕ $(3+4i)(2-5i)$

$26-7i$

㉖ $(-6+2i)(3+8i)$

$-34-42i$

㉗ $(9-3i)(8-2i)$

$66-42i$

㉘ $(-1+2i)(-1+10i)$

$-19-12i$

㉙ $(4+5i)^2$

$-9+40i$

㉚ $(1-7i)^2$

$-48-14i$

Write the following expressions in $a+bi$ form.

㉛ $\frac{8}{2+i}$

$\frac{16}{5}-\frac{8}{5}i$

$$(32) \frac{-7}{3-4i}$$

$$\boxed{\frac{-21}{25} - \frac{28}{25}i}$$

$$(33) \frac{-2}{-3+7i}$$

$$\frac{6}{58} + \frac{14}{58}i = \boxed{\frac{3}{29} + \frac{7}{29}i}$$

$$(34) \frac{2-i}{2+i}$$

$$\boxed{\frac{3-4}{5} - \frac{4}{5}i}$$

$$(35) \frac{3+2i}{-4+3i}$$

$$\boxed{\frac{-6}{25} - \frac{17}{25}i}$$

$$(36) \frac{1-8i}{5-6i}$$

$$\boxed{\frac{53}{61} - \frac{34}{61}i}$$

$$(37) \frac{2-3i}{4i}$$

$$\boxed{-\frac{3}{4} - \frac{1}{2}i}$$

$$(38) \frac{5+5i}{-3i}$$

$$\boxed{-\frac{5}{3} + \frac{5}{3}i}$$

Simplify.

$$(39) i^{30}$$

$$\boxed{-1}$$

$$(40) i^{71}$$

$$\boxed{-i}$$

$$(41) i^{42}$$

$$\boxed{-1}$$

$$(42) i^{93}$$

$$\boxed{i}$$

$$(43) i^{60}$$

$$\boxed{1}$$

Write each expression in terms of i .

$$(44) \sqrt{-6}$$

$$\boxed{i\sqrt{6}}$$

$$(45) \sqrt{-25}$$

$$\boxed{5i}$$

$$(46) \sqrt{-9}$$

$$\boxed{3i}$$

$$(47) \sqrt{-10}$$

$$\boxed{i\sqrt{10}}$$

$$(48) \sqrt{-36}$$

$$\boxed{6i}$$

$$(49) \sqrt{-13}$$

$$\boxed{i\sqrt{13}}$$

$$(50) -2\sqrt{-63} + 3\sqrt{-28}$$

$$\boxed{0}$$

$$(51) 7\sqrt{-44} - 8\sqrt{-99}$$

$$\boxed{-10i\sqrt{11}}$$

$$(52) -6\sqrt{-18} - 9\sqrt{-2}$$

$$\boxed{-27i\sqrt{2}}$$